

Store Footprint Efficiency Analysis

SQL Exploration Report

Business Problem: The company runs 66 physical stores across multiple countries. Each store occupies a different amount of floor space, but it's unclear if that space is used efficiently.

Goal: Find revenue generated per square meter for each store, rank stores best to worst, and identify underperformers so leadership can decide where to expand, shrink, or close store space.

Section 1: Base Data Check

Q: How many physical stores exist, and do all have valid Square Meters values?

SQL Query

```
SELECT
    COUNT(*) AS [Total Stores],
    SUM(CASE WHEN Square_Meters IS NULL OR Square_Meters = 0
        THEN 1 ELSE 0 END) AS [Invalid Square Meters Value]
FROM dbo.Stores
WHERE dbo.Stores.StoreKey <> 0;
```

Result

Total Stores	Invalid Square Meters
66	0

Insight: 66 physical stores in total, all with valid Square Meters. Only StoreKey 0 (Online) is excluded.

Section 2: Store-Level Revenue

Q: What is the total revenue generated by each physical store, including stores with zero sales?

SQL Query

```
SELECT St.StoreKey, St.Country, St.Square_Meters, St.Open_Date,
    COALESCE(SUM(S.Quantity * P.Unit_Price_USD), 0) AS [Total Revenue]
FROM Stores St
LEFT JOIN Sales S ON St.StoreKey = S.StoreKey
LEFT JOIN Products P ON S.ProductKey = P.ProductKey
WHERE St.StoreKey <> 0
GROUP BY St.StoreKey, St.Country, St.State, St.Square_Meters, St.Open_Date
ORDER BY [Total Revenue] DESC;
```

Result

StoreKey	Country	Sqm	Total Revenue
66	United States	840	\$1,221,970
9	Canada	1,190	\$1,060,020
...	...	...	...
52	United States	2,000	\$0
46	United States	1,015	\$0

Insight: 9 out of 66 stores made zero revenue, despite having decent floor space. These are the biggest red flags in the whole dataset.

Section 4: Country-Level Rollup

Q: Is underperformance concentrated in specific countries, or spread across all of them?

SQL Query

```
SELECT Country, COUNT(StoreKey) AS NumStores,
       SUM([Total Revenue]) AS CountryRevenue, SUM(Square_Meters) AS CountrySqm,
       ROUND(SUM([Total Revenue]) / NULLIF(SUM(Square_Meters), 0), 2) AS AvgRevenuePerSqm
FROM StoreRevenue
GROUP BY Country
ORDER BY AvgRevenuePerSqm DESC;
```

Result

Country	Stores	Dead Stores	Avg Rev/Sqm
United States	24	4	\$625.54
France	7	0	\$514.45
Canada	5	2	\$506.53
Italy	3	0	\$478.86
United Kingdom	7	0	\$456.33
Germany	9	1	\$333.04
Netherlands	5	1	\$259.81
Australia	6	1	\$226.69

Insight: US looks best on paper but has the most dead stores (4) — hidden by scale (24 stores total). Netherlands and Australia look worst only because they have few stores, so 1 dead store drags the average down hard.

Section 5: Store Size Segmentation

Q: Do small, medium, and large stores differ in average revenue-per-sqm?

SQL Query

```
SELECT [Size Bucket], COUNT(StoreKey) AS [Total Stores],
       ROUND(AVG([Total Revenue] / NULLIF(Square_Meters, 0)), 2) AS AvgRevenuePerSqm
FROM Sized
GROUP BY [Size Bucket]
ORDER BY AvgRevenuePerSqm DESC;
```

Result

Size Bucket	Store Count	Avg Rev/Sqm
Small (<1000 sqm)	13	\$575.32
Medium (1000-2000)	48	\$484.90
Large (>2000 sqm)	5	\$410.15

Insight: Small stores are most productive per sqm. Bigger floor space is not paying off.

Section 6: Store Age Check

Q: Are underperforming stores mostly newer (ramping up) or older (genuine problem)?

SQL Query

```
SELECT YEAR(Open_Date) AS OpenYear, COUNT(StoreKey) AS NumStores,  
       ROUND(AVG(TotalRevenue / NULLIF(Square_Meters, 0)), 2) AS AvgRevenuePerSqm  
FROM StoreRevenue  
GROUP BY YEAR(Open_Date)  
ORDER BY OpenYear;
```

Result

Open Year	Store Count	Avg Rev/Sqm
2005	3	\$430.57
2007	5	\$223.08
2008	8	\$445.49
2009	6	\$317.18
2010	11	\$493.75
2012	14	\$475.24
2013	3	\$496.88
2014	3	\$894.42
2015	9	\$676.89
2018	3	\$593.68
2019	1	\$799.36

Insight: 2007 and 2009 were clearly weak vintages. Other old stores (2005, 2010, 2012, 2013) are healthy — it's specific batches, not age itself.

Section 7: Outlier Flagging

Q: Which stores deviate significantly from their country average revenue-per-sqm?

SQL Query

```
SELECT StoreKey, Country, Square_Meters, RevenuePerSqm,  
       CountryAvgRevPerSqm, DeviationFromCountryAvg  
FROM WithCountryAvg  
ORDER BY DeviationFromCountryAvg ASC;
```

Result

StoreKey	Country	Rev/Sqm	Country Avg	Deviation
52	United States	\$0.00	\$648.40	-\$648.40
46	United States	\$0.00	\$648.40	-\$648.40
...	...	...	...	...
9	Canada	\$890.77	\$467.66	+\$423.11
66	United States	\$1,453.53	\$648.40	+\$805.13

Insight: Best: Store 66 (US) beats its country average by \$805/sqm. Worst: the 4 dead US stores fall hardest in dollar terms since the US average is high — dead stores in weaker countries show smaller dollar gaps, but the failure is just as real.

Final Insights

1. 9 of 66 stores (14%) generate zero revenue — dead floor space across 5 countries, a company-wide issue.
2. Country-level averages are misleading — they hide problems in large markets (US) and exaggerate them in small ones (Netherlands, Australia).
3. Small stores are more efficient per sqm than large stores — bigger footprint isn't translating to better returns.
4. Store performance is tied to when it was opened — 2007 and 2009 were weak vintages, 2014 and 2019 were strong.
5. Strong performance is achievable at scale — Store 66 proves it, beating its country average by over \$800/sqm.

Business Recommendations

1. Audit and act on the 9 zero-revenue stores first — relocate, close, or investigate local causes.
2. Freeze new large-format store openings until the size-vs-productivity gap is understood.
3. Investigate the 2007/2009 store cohort specifically — bad expansion strategy or site selection?
4. Study Store 66 as a best-practice case and replicate what it's doing differently.
5. Rebalance floor space — consider downsizing large underperforming stores rather than closing outright.